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# Famous Astronomers: Subrahmanyan Chandrasekhar

The Limit of Stars

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## Introduction

Some scientists add a brick to the edifice; a rare few lay a foundation stone that the whole structure quietly rests on. Subrahmanyan Chandrasekhar, the mathematician-physicist known to colleagues simply as Chandra, belonged to the second kind. At nineteen, on a passenger ship crossing from India to England, he worked out a calculation that told humanity there is a strict weight limit to the afterlife of a star, and that a dying sun heavier than that limit must collapse into something far stranger than anyone then imagined. The idea was ridiculed by the most famous astronomer alive. Half a century later it was a pillar of modern astrophysics, and its author stood in Stockholm.

This volume tells his story in full: the prodigy from Madras raised in the orbit of a Nobel-awarding uncle, the ocean voyage that produced a landmark of physics, the public humiliation in London that scarred and steeled him, the long American decades at the University of Chicago during which he remade one field of theoretical astronomy after another, and the final vindication that arrived with the discovery of the very objects his youthful equations had predicted.

The book belongs to the Famous Astronomers series, which walks through the lives and legacies of the people who built humanity's picture of the cosmos. Chandra's chapter of that story is unusual in two ways. His greatest discovery was made before he held any degree at all, and it was rejected rather than celebrated for decades, which makes his life a study in how science actually absorbs unwelcome truths. And his career was not a single quest but a sequence of complete careers, each spent mastering a subject, writing its definitive book, and moving deliberately on to the next.

Everything here is evergreen: the physics has not aged, the history is settled, and the man's example, that style and patience are forces in science, remains as fresh as the latest gravitational wave detection that leans on his mathematics. Readers who want a human story will find one of the strangest and most moving in the history of physics. Readers who want the science will find the clearest explanation of his limit and its consequences that these pages can honestly give.

The long game is the frame, and the introduction should set it before the chapters play it out. Here is a career with an arc unlike the usual scientific biography: the founding result computed at nineteen, the first public verdict a humiliation, the vindication a half century in the making, and the Nobel medal arriving when the laureate was past seventy; the reader should hold the arc in view from the first page, because the individual chapters only make sense as movements of one long composition, the man's own metaphor for his science being music and the biography reading as a performance with an unusually slow tempo, the introduction promising that the slow tempo is the story, the long game itself being the lesson this life keeps teaching.

A note on the name, because the reader will meet it in several forms before the book is done. The subject of this volume appears in the records as a long South Indian name, in conversation and among colleagues as the single syllable his friends used, and in the present sky as the name of a spacecraft; the account will use the full name in the chapters' openings and the friendly form where the narrative warms, following the memoirs' own practice, and the reader is invited to note the rare honor the third form represents: a man whose professional name became an observatory in orbit, the three forms marking the three scales of the life, the person, the colleague, and the institution.

## Chapter 1: The Discovery of Subrahmanyan Chandrasekhar

The man who set a weight limit on dead stars was born into a family that already treasured exact science. His birth in 1910 came in Lahore, then part of British India, into a Tamil Brahmin household of civil servants and scholars; his father held a post in the accounts service, and music and mathematics were the household's twin languages. The uncle who mattered most was his father's brother, Chandrasekhara Venkata Raman, a physicist of mounting reputation who would win the Nobel Prize in 1930 for the scattering effect that carries his name. The boy grew up half in his uncle's orbit, watching a working scientist from inside the family.

His schooling was Madras at its best, with tutors and a strong school, and the child was a genuine prodigy with numbers. By his mid-teens he was reading university-level physics on his own, teaching himself the new quantum theory from the papers arriving in the scientific journals, work produced by men barely a decade older than he was. At fifteen he crossed paths with a founder of the field itself: Arnold Sommerfeld, the Munich physicist whose atomic theory was then famous, visited Madras on a lecture tour, and the boy sought him out to discuss the new

mechanics. Sommerfeld gave him the proofs of an upcoming paper, the mathematical gear of the new quantum statistics, and the encounter convinced the teenager that he could work on the frontier himself rather than read about it later.

He enrolled at Presidency College in Madras and began producing publishable research almost immediately. His first scientific paper, on the quantum treatment of a classic problem in the theory of light's interaction with matter, appeared while he was still an undergraduate, and more followed. The subjects he touched, the absorption of light by atoms and the behavior of particles obeying the new statistics, were precisely the tools he would soon turn on the stars. He was, by every account, the most gifted physics student in India, and he had the luck of coming of age at the exact moment when quantum mechanics made the physics of matter-under-extreme-pressure a solvable problem.

The voyage that made him famous began in the summer of 1930. He had won a scholarship from the government of India to study in England, and he boarded the steamer at Bombay bound for Cambridge. The sea passage lasted weeks, and Chandra, aged nineteen, carried with him a copy of a landmark paper by the Cambridge theorist Ralph Fowler on the dense stellar corpses then just being identified. Fowler had shown that the dead remnant of a star, a white dwarf, is held up not by heat but by a quantum effect: the packed electrons refuse, by the new statistics, to be squeezed into the same states, and their refusal produces the pressure that keeps the corpse inflated. Chandra wondered what would happen if the star were heavier, and he spent the crossing doing the arithmetic. His improvement was to combine Fowler's quantum pressure with a second, older piece of physics, Einstein's relativity, which forces the fastest electrons toward the speed of light, where they grow heavier and their pressure weaker.

The answer he carried off the ship was astonishing: a white dwarf more massive than a critical weight simply cannot hold itself up at all, and no stable dead star exists above that weight. He arrived in England carrying the first draft of a result that would take twenty years to be believed and fifty to be crowned with a Nobel Prize.

His English student years deserve a closing paragraph before the profile begins. At Cambridge he studied under the very men whose papers he had carried across the ocean, passed the mathematical examinations with the highest honors of his college, won its prize fellowship, and was elected to the Royal Astronomical Society while still a research student, a rise so rapid that older members assumed he must be someone's protg, which in a sense he was: the whole quantum generation had unknowingly prepared him. He spent summers visiting the great institutes of Europe, working in the Copenhagen atmosphere where the new physics was being argued into shape, and he presented his white dwarf papers across the continent to audiences that received them politely and filed them quietly. The continent's physicists, unlike England's astronomers, had no difficulty with his arithmetic, and their private support, recorded in letters, kept his confidence intact through the years ahead.

A note on his tools completes the discovery chapter, since the image matters. He carried onto the ship a knowledge of the new statistics learned from a founder's own lecture notes, a supply of ruled paper, and the ocean's worth of uninterrupted time; the calculation that resulted was done with the instruments of the era, meaning patience, arithmetic, and the compact notation of thermodynamics. No computer has ever been involved in the original derivation of the limit, and the physics he wrote down then has never needed to be redone, only extended; the modern reader who checks it finds the same number, which is the whole point of doing mathematics correctly in the first place.

The uncle's shadow is part of the discovery, because the household's science had a Nobel tone before the child could read. The boy's uncle, his mother's brother, was the era's rising physicist of the subcontinent, the experimentalist of light scattering whose own medal would arrive while the nephew was still a teenager; the family's letters and visits kept physics in the air of the childhood, the uncle's example a standing proof that the world's highest scientific honors could have an address in India; the chapter's honest reading: the first discovery about the boy was that science was normal in his house, the profession not a foreign ornament but a family trade, and the uncle's parallel rise giving the nephew a horizon within sight, the household's two careers eventually bracketing the century of Indian physics, the uncle's prize the first, the nephew's the second for the family.

The city of his formation earns a passage, because the young man's Madras did more than school him. The presidency college of the coastal metropolis gave the teenager the mathematics of a university while he was still officially its junior, the local scientific society of the era opening its meetings to the precocious student, the first papers written from home and published in the foreign journals before the author had a degree; the chapter's honest portrait: the colonial city's intellectual life was thin by the metropole's standards and yet sufficient for a genius, the libraries and the societies and the mail carrying the teenager into the world's conversation, the

discovery chapter keeping the city because the man was made there, the future arriving in Madras as journals and correspondence before it arrived as a career.